

03 · Price elasticity by region — random slopes & endogeneity

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The business decision. We set prices region by region and want each region’s **price elasticity of demand** — the standard economics measure of how sensitive demand is to price. Concretely, elasticity is the **% change in quantity sold for a 1% change in price**; it’s almost always negative (raise price → sell less). An elasticity of -1.5 means “cut price 1%, sell 1.5% more.” We need it per region because a tourist market and a price-sensitive suburb should not be priced the same.

Two traps this notebook is built around

1. **Small regions are noisy.** A region with 12 weeks of data will hand you a wild elasticity estimate that’s mostly sampling noise. Acting on it is overfitting. The fix is **partial pooling** (below).
2. **Price is endogenous.** “Endogenous” means the treatment (price) is *correlated with the error term* — here because managers historically *cut prices when they expected soft demand*. So low prices coincide with low demand for reasons that have nothing to do with the price cut, which drags the naive elasticity estimate *toward zero* (makes demand look less price-sensitive than it is). No amount of clever averaging fixes this — it needs an **instrument** (notebook 11). Step 0 meets it in classical form, before any prior appears; step 8 breaks the assumption on purpose and prices the cure.

The tools, in plain terms

- **Log-log demand.** We model $\log(\text{quantity}) = \alpha + \beta \log(\text{price}) + \dots$. The magic of taking logs on both sides is that the slope β **is** the elasticity directly (a constant %-for-% relationship), so there’s one interpretable number per region. One line of calculus shows why: elasticity is defined as $\epsilon(P) \equiv \frac{dQ}{dP} \cdot \frac{P}{Q}$, and for the constant-elasticity curve $Q = AP^\beta$ we get $\epsilon(P) = \beta AP^{\beta-1} \cdot \frac{P}{AP^\beta} = \beta$ — the *same* β at every price point. For **any other** demand shape the elasticity moves with the price level, so β is best read as a *local approximation around the prices we actually observed* — a point that returns with teeth when we recommend prices outside that range.
- **Random slopes / partial pooling.** Instead of one elasticity for all regions (ignores real differences) or a separate free estimate per region (noisy), we assume each region’s elasticity β_r is drawn from a shared distribution $\beta_r \sim \mathcal{N}(\mu_\beta, \tau_\beta^2)$. This **hierarchical** structure makes each region’s estimate a compromise between its own noisy data and the fleet average — data-poor regions get pulled (“shrunk”) harder toward the fleet, which is exactly the protection we want. μ_β is the typical elasticity; τ_β measures how much regions genuinely differ.

Cookbook maps this to **pathmc** random slopes; here we fit the equivalent hierarchical model explicitly in PyMC so the shrinkage is fully visible. The identification logic is

unchanged.

On real data. Swap in your **own transaction panel** — one row per region per week with price, units sold, and controls (competitor price, promotions, seasonality). Public analogues are retail *scanner* datasets (e.g. the Dominick’s Finer Foods store data). The endogeneity caveat is not academic: it’s the single biggest reason naive “price tests” mislead in practice.

2 · Simulate a ground truth

A region $\times$ week panel with a **constant-elasticity (log-log) demand curve** $\log Q_{rt} = \alpha_r + \beta_r \log P_{rt} + \gamma^\top Z_{rt} + \varepsilon_{rt}$, where β_r is region r ’s true price elasticity (planted, varying around a fleet mean ≈ -1.4). Of the observed covariates Z , only **competitor price** is a genuine confounder — it moves both our price and demand; **season** and **trend** shift demand only, so they enter as *precision covariates* (they sharpen estimates), not confounders. We start with **no** endogeneity so we can show recovery, then switch it on in step 8.

The data-generating model — exactly what `dgp.price_panel` implements (defaults & seed in `src/cmp/dgp.py`). For region r , week t : season $s_t = 0.3 \sin(2\pi t/26)$, trend $g_t = 0.01 t$, competitor price $C_{rt} \sim \mathcal{N}(20, 2^2)$, and a demand shock $u_{rt} \sim \mathcal{N}(0, 1)$:

$$\begin{aligned}\beta_r &\sim \mathcal{N}(-1.4, 0.35^2) \\ \log P_{rt} &= \log 15 - 0.05 C_{rt}/20 + \kappa u_{rt} + \nu_{rt}, \quad \nu_{rt} \sim \mathcal{N}(0, 0.05^2) \\ \log Q_{rt} &= 2.5 + \beta_r \log P_{rt} + 0.4 \frac{C_{rt}-20}{20} + s_t + g_t + u_{rt} + \varepsilon_{rt}, \quad \varepsilon_{rt} \sim \mathcal{N}(0, 0.08^2).\end{aligned}$$

Line 1 plants the true regional elasticities; line 2 is the price-setting rule; line 3 is demand. C_{rt} appears in **both** equations — that is what makes competitor price the genuine confounder (it moves our price *and* demand), while s_t and g_t appear only in the demand equation (precision covariates). κ is the **endogeneity switch**: through step 6 the generator runs with $\kappa = 0$ and zeroes the shock entirely ($u_{rt} \equiv 0$), so recovery is clean; step 8 sets $\kappa > 0$, and the *same* shock now enters both equations — managers’ prices respond to the very shock that moves demand ($\text{Cov}(\log P, u) > 0$), the simultaneity that biases the naive elasticity toward zero.

12 regions, fleet-mean elasticity -1.42; obs per region range 16-80 (regions 0-2 are thin, to exercise shrinkage)

	region	week	price	demand	competitor_price	season \
0	region_00	2	13.003239	0.159630	20.405765	1.394170e-01
1	region_00	3	14.245949	0.157405	16.535730	1.989368e-01
2	region_00	12	14.310937	0.159097	20.819276	7.179470e-02
3	region_00	13	15.626376	0.175255	21.659711	-9.648736e-17
4	region_00	15	13.204126	0.152195	19.486540	-1.394170e-01

	trend	log_price	log_demand
0	0.02	2.565199	-1.834900
1	0.03	2.656473	-1.848934
2	0.12	2.661024	-1.838239
3	0.13	2.748960	-1.741513

4 0.15 2.580529 -1.882592

3 · Identify — what must be true for β_r to be causal?

Identification vs estimation. *Identification* asks whether the causal quantity could be recovered from this data even with infinite samples — a question about **assumptions**, settled before any fitting; *estimation* is the statistics we do once identification is granted. Partial pooling (below) is an **estimation** device — it buys variance reduction, never identification.

Estimand. Region r 's elasticity is a *causal* slope — what happens to demand when **we** move the price, not when the market moves it for us:

$$\beta_r = \frac{\partial \mathbb{E}[\log Q \mid \text{do}(\log P = p), r]}{\partial p},$$

where $\text{do}(\cdot)$ marks an *intervention* on price (Pearl's do-operator), as opposed to passively observing whatever prices managers happened to choose. The regression slope equals this causal slope only under assumptions — four of them, each named, formalized, and honestly labelled as checkable or not:

A1 · Conditional price exogeneity (the continuous-treatment version of **unconfoundedness** — once we hold the observed controls fixed, price is as-good-as-randomly assigned with respect to the demand shock):

$$\varepsilon_{rt} \perp \log P_{rt} \mid Z_{rt}, r.$$

After conditioning on region and the observed controls Z , whatever noise remains in demand must be unrelated to how the price was set. The one true observable confounder in our DGP is **competitor price** — it moves both P and Q , opening a **backdoor path** in the DAG below — a *backdoor path* is a non-causal route linking price and demand through a common cause (here $P \leftarrow \text{competitor price} \rightarrow Q$) that manufactures correlation which is *not* the price effect — and conditioning on it closes that path. **Season** and **trend** move demand only, so they are *precision covariates* (they sharpen estimates), not confounders. A1 holds by construction while $\kappa = 0$; it is *exactly* the assumption we break in step 8, and it is **untestable from the data alone** — no diagnostic in this notebook can certify it.

A2 · Price variation — genuine price movement left after the controls (the continuous-treatment form of **positivity**):

$$\text{Var}(\log P_{rt} \mid Z_{rt}, r) > 0.$$

Each region must have genuine price movement left over after the controls; no variation, no slope. In the DGP this is the quasi-experimental wiggle $\nu_{rt} \sim \mathcal{N}(0, 0.05^2)$ — checked numerically in the next cell. In real data this is the assumption most often *silently* violated: uniform corporate pricing or rigid cost-plus rules leave $\text{Var}(\log P \mid Z) \approx 0$, and the “elasticity” you then estimate is noise dressed up as a number.

A3 · No dynamic effects / SUTVA (Stable Unit Treatment Value Assumption). This week’s price affects this week’s demand only: no stockpiling (“pantry-loading” ahead of a price rise), no delayed reference-price effects, and no cross-region spillovers (region A’s price doesn’t move region B’s demand — no arbitrage). Our DGP satisfies this by construction; on real scanner data a `lag(price)` term is the standard first probe (step 8’s caveats).

A4 · Functional form. In a *parametric* model the constant-elasticity (log-log) shape is part of the identification bundle: if the true curve bends, the “constant” β_r is a price-range-specific average. We stress-test this in-sample in step 5 with a curvature term — but no in-sample check reaches prices we never charged.

Partial pooling, and what it buys. The estimand is per-region, so we treat elasticity as a **random slope**: $\beta_r \sim \mathcal{N}(\mu_\beta, \tau_\beta^2)$ — noisy regions shrink toward the fleet mean μ_β , and τ_β (a standard deviation — the between-region spread of elasticities) *measures how much elasticity genuinely varies*. The **shrinkage factor** for region r is approximately $1 - \frac{\tau_\beta^2}{\tau_\beta^2 + \sigma^2/n_r}$ — thin regions (small n_r) get pulled harder toward the fleet. Shrinkage is a feature: it stops us over-reacting to a small region’s noisy slope. But pooling handles *noise*, not *bias*: if A1 fails, all three pooling regimes in step 4 are biased by the same amount.

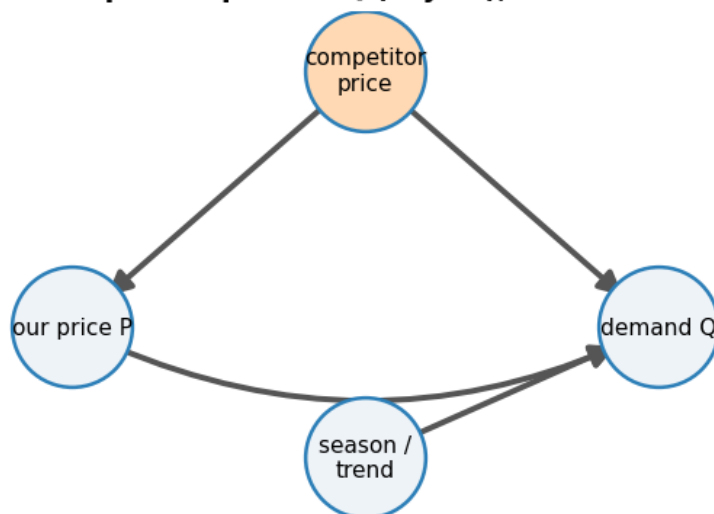
How big is the bias if A1 fails? Two lines of algebra, not hand-waving. Suppose an unobserved shock u_{rt} (variance σ_u^2) enters demand with coefficient 1 *and* leaks into price with coefficient κ (managers cut price when they expect soft demand). For the regression of $\log Q$ on $\log P$ after partialling out Z and region effects (i.e. after regressing price and demand on the controls and keeping only the residual variation), the classic **omitted-variable-bias** formula gives

$$\hat{\beta} \xrightarrow{p} \beta + \frac{\text{Cov}(\log P, u)}{\text{Var}(\log P)} = \beta + \frac{\kappa \sigma_u^2}{\kappa^2 \sigma_u^2 + \sigma_\nu^2},$$

using the DGP’s price equation ($\text{Cov}(\log P, u) = \kappa \sigma_u^2$ and $\text{Var}(\log P | Z) = \kappa^2 \sigma_u^2 + \sigma_\nu^2$). The arrow $\xrightarrow{p}$ reads “converges in probability to” — the value the estimate settles on as the sample grows. The bias term is **positive**, and since $\beta < 0$ it drags $\hat{\beta}$ **toward zero** — demand looks less price-sensitive than it is. Worse, the *sign of the whole slope can flip*: when the exogenous wiggle σ_ν is small, even a faint κ makes the bias enormous (the denominator is tiny) and the fitted “demand curve” slopes *upward* — high prices coinciding with high demand because both chase the same shock. Step 8 verifies this arithmetic numerically, then sweeps κ to map the whole curve. What pooling does **not** fix is exactly this: an *unobserved* common shock needs an instrument — previewed in step 8, full treatment in notebook 11. We state this out loud rather than hide it.

A2 check - within-region SD of log-price: min 0.046, median 0.052, max 0.058
(all comfortably > 0 : positivity holds, by design)

Backdoor $P \leftarrow \text{competitor price} \rightarrow Q$ (adjust); season/trend = precision



Step 0 · The classical read (no likelihood, no priors, no sampler)

Before any Bayesian machinery, ask what a competent analyst does here **without** it. The answer is not a different analysis: it is the *same estimand* (§3’s causal slope), identified by the *same four assumptions* (A1–A4), estimated with the simplest tool that is still correct. And in a log-log demand curve that tool is almost embarrassingly simple, because **the estimand is a regression coefficient**:

$$\log Q_{rt} = \alpha_r + \beta \log P_{rt} + \gamma^\top Z_{rt} + \varepsilon_{rt},$$

with region intercepts α_r (twelve dummies) and Z the §3 adjustment set (competitor price, season, trend). The coefficient on $\log P$ is the elasticity — no back-transformation, no marginal effect to compute, no delta method. One `cl.ols` call returns it, and it is the number the whole notebook is about.

What the classical arm cannot do — and this is the crux. The estimand is per-region, and OLS has exactly **two** ways to deliver a per-region elasticity: pretend the regions are identical (**complete pooling** — the single β above, one number quoted twelve times) or estimate each region from its own weeks alone (**no pooling** — twelve separate regressions, three of them on 16 observations). There is no third option in the classical toolbox. Hold that thought: §4’s hierarchical model is the third option, and **5x** settles the account between them.

The covariance choice is not a detail — it is the estimate’s other half. Our region-weeks are not independent observations, and for two reasons, of which only the second is the textbook one:

1. **The pooled model is deliberately wrong about slopes.** §2 planted $\beta_r \sim \mathcal{N}(-1.4, 0.35^2)$ — regions genuinely differ. Force one shared β on them and each region’s residual inherits

a term $(\beta_r - \beta) \log P_{rt}$ that is *the same sign, week after week, for that region*. That is within-region correlation manufactured by the specification itself.

2. **Real panels add serially-correlated local shocks** on top (a soft month tends to be followed by another soft month).

Neither is visible to the iid or HC1 formulas, which only ever look at one row at a time. **Clustering on region** lets the errors be arbitrarily correlated *within* a region and asks only for independence *across* regions, so the **12 regions**, not the 768 rows, become the unit of inference — the correction of **Bertrand–Duflo–Mullainathan (2004)**, whose famous finding is that ignoring within-panel correlation can leave standard errors off by a factor of five or more. So we fit the same regression **three times** — iid, HC1, region-clustered — and print all three. The point estimate will be identical to the last decimal; only the story about its precision changes.

One caveat to hear *before* the number appears. We have **12** clusters. Cluster-robust standard errors are a large- G approximation; at $G = 12$ they are themselves *downward*-biased (the usual rule of thumb wants 30–50 clusters, and with fewer you are supposed to reach for a wild cluster bootstrap). So the clustered interval below is the best interval the classical toolbox has here — and it is still, mildly, an optimistic one. No covariance option fixes that. The honest fix is to *model* the region-level variation, which is exactly what §4 will do.

The confounding story, priced. §3 claimed competitor price is the one genuine confounder and that season/trend are mere *precision covariates*. We test that claim by dropping controls one block at a time. And because a single sample cannot tell **bias** from **bad luck**, we then repeat the whole exercise on fresh panels and report the *average* error across seeds — the only way to see which omissions move the estimate systematically and which merely make it wobble.

THE ESTIMAND - the fleet price elasticity: the log-price coefficient of the pooled log-log fit (768 region-weeks, 12 regions).

One point estimate; three stories about how precise it is:

iid (nonrobust)	-1.39	[90% CI -1.49, -1.30]	SE 0.058
HC1 (robust)	-1.39	[90% CI -1.49, -1.29]	SE 0.061
clustered on region	-1.39	[90% CI -1.54, -1.24]	SE 0.091

-> clustering on region multiplies the SE by x1.50 vs HC1 (x1.58 vs iid).

GRADE vs the planted truth (fleet-mean beta = -1.42): the pooled estimate is off by +0.03, and the truth falls **INSIDE** the clustered 90% CI. The classical estimator lands the **FLEET** number; what it cannot land is the twelve regional ones (5x).

THE CONTROL LADDER - competitor price is the confounder, season/trend are the precision covariates (region-clustered SEs throughout):

all controls (section 3's)	-1.39	[90% CI -1.54, -1.24]	SE 0.091
drop COMPETITOR PRICE	-1.44	[90% CI -1.60, -1.27]	SE 0.102
drop SEASON + TREND	-1.64	[90% CI -1.90, -1.38]	SE 0.158
drop everything (bivariate)	-1.52	[90% CI -2.23, -0.81]	SE 0.431

BIAS vs NOISE - the same ladder over 25 fresh panels (mean error against each panel's own planted truth):

full	mean error +0.016	(seed-to-seed sd 0.043)
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no competitor      mean error -0.062   (seed-to-seed sd 0.040)
no season/trend    mean error +0.043   (seed-to-seed sd 0.175)
typical SE: full 0.054 vs no season/trend 0.186 (x3.4 wider)

```

Read the three covariance rows, then the ladder.

The covariance. Clustering on region *widened* the interval — the ratio is printed above. That is the direction people expect, but not for the reason they expect: our DGP’s weekly shocks are iid by construction, so there is no serial correlation for the cluster-robust sandwich to find. What it finds is the *first* mechanism listed above — the pooled model’s own misspecification. Each region’s residuals carry that region’s slope deviation $(\beta_r - \beta) \log P_{rt}$, week after week, and HC1 (which inspects one row at a time) cannot see it, so it reports a precision the pooled model does not have. Clustering looks at the twelve regional trajectories, sees the structure, and charges for it. **The iid and HC1 intervals are not conservative or anti-conservative as a rule — they are simply wrong in a direction you cannot predict without doing the clustering.** And note the tell: the widening is itself evidence that the pooled specification is misspecified. The classical machinery is *telling us* that one β is not enough — it just has no way to act on the message.

The ladder. Two omissions, two entirely different failures, and only the seed loop can tell them apart:

- Dropping **competitor price** — the real confounder, the one with the backdoor path in the DAG — moves the estimate by a *small but systematic* amount: across fresh panels the mean error is clearly non-zero (printed above), and it points **away from zero** (demand looks *more* price-sensitive than it is). The sign is exactly what the DGP dictates: a competitor price rise *lifts* our demand (+0.4) while *lowering* our own price ($-0.05 C/20$), so an analyst who ignores it sees cheap weeks coinciding with strong weeks and credits the price cut for demand the competitor delivered. That is textbook omitted-variable bias, and unlike noise it does **not** shrink with more data. It is small here only because that passthrough is weak by construction; with a strong one the same mechanism is ruinous. (Note it is a *different* bias from §3’s toward-zero formula, which concerns the **unobserved** shock u — that one arrives in §8, and it is an order of magnitude worse.)
- Dropping **season and trend** — the “precision covariates”, which the DAG says are *not* confounders — leaves the mean error near zero across seeds (no bias, exactly as advertised) but **multiplies the standard error several-fold** and inflates the seed-to-seed scatter to match. On *this* particular panel it happened to throw the point estimate a long way (see the ladder print) — and that is *noise*, not bias, which is precisely what a single sample can never tell you and the seed loop can.

The practical lesson generalizes past this notebook: when the treatment barely moves — here $\log P$ has a standard deviation of about 0.05 — anything that explains a large share of the *outcome* is worth controlling for even when it is not a confounder, because the residual variance it removes is what the slope is estimated from. “Precision covariate” sounds optional. It is not.

The assumption that will break — and the classical cure, met now

A1 (conditional price exogeneity) is the load-bearing assumption, and §8 breaks it on purpose: managers cut prices *because* they expect soft demand, so an **unobserved** shock u_{rt} sits in the price equation and in the demand equation at once. No standard error fixes that, no amount of pooling

fixes that, and — the point worth internalizing — **no diagnostic in §5 will even notice**. The classical world has a cure, and the reader should meet it *before* meeting any prior, because it is not a Bayesian idea at all.

The cure is an **instrument** W : a variable that moves **price** but has **no path to demand except through price**. The textbook pricing instrument is a **cost shifter** — a wholesale-cost or exchange-rate shock moves what we charge, but shoppers never see it except in the price tag. Three conditions, and only the first is checkable:

- **Relevance:** $\text{Cov}(W, \log P) \neq 0$ — measured by the **first-stage** F , i.e. how hard the instrument moves price once the controls are partialled out (Stock–Yogo rule of thumb: $F < 10$ means the instrument is too weak to use, and a weak instrument is *worse* than none — it drags 2SLS back toward OLS while inflating its interval).
- **Exclusion:** W affects $\log Q$ *only through* $\log P$ — untestable, argued from institutions (customers do not observe our wholesale costs).
- **Independence:** $W \perp u$ — cost shocks do not chase demand shocks.

Two-stage least squares (2SLS) then throws away every scrap of price variation except the clean, W -driven slice:

$$\text{stage 1: } \widehat{\log P} = \hat{\pi}_0 + \hat{\pi}_1 W + \hat{\pi}_Z^\top Z, \quad \text{stage 2: } \log Q = \alpha + \beta_{\text{IV}} \widehat{\log P} + \gamma^\top Z + e.$$

The endogenous slice of price (κu) is orthogonal to W , so stage 1 filters it out and β_{IV} recovers the causal slope. `dgp.price_panel` has no instrument built in, so we plant one on the $\kappa = 1.2$ panel by hand: a cost shock $W_{rt} \sim \mathcal{N}(0, 1)$ that shifts log-price by $0.25 W_{rt}$ and shifts log-demand **only through that price move** (by $\beta_r \times$ the price shift) — exclusion true by construction. `cl.iv_2sls` does the algebra, with the *proper* 2SLS standard errors (not the naive second-stage OLS output, which ignores that stage 1 was itself estimated) and the first-stage F . This panel and these two fits are built **once, here**, and §8 reuses them.

A1 BROKEN ($\kappa = 1.2$). Truth is unchanged: fleet-mean $\beta = -1.42$.

```
OLS (endogenous)      -0.63  [90% CI -0.78, -0.48]  SE 0.092
2SLS (instrumented)   -1.66  [90% CI -2.01, -1.31]  SE 0.212
first-stage F = 27 (>10: relevant, not weak)
-> OLS : 8.6 SEs from the truth; the truth is OUTSIDE its 90% CI.
-> 2SLS: 1.1 SEs from the truth; the truth is INSIDE its 90% CI.
```

OLS is PRECISELY WRONG — and its most 'efficient' standard error is the most confident lie of all: under HC1 the same estimate carries SE 0.02 (vs 0.09

clustered), putting the truth 45 SEs away.

2SLS is IMPRECISELY RIGHT — its interval is x2.3 wider, and it covers.

(`cl.iv_2sls` reports homoskedastic 2SLS SEs; clustering them would widen it further, so even this is the optimistic version of the IV bargain.)

THE DECISION, not the decimal: section 7's pricing rule acts only where $\beta < -1$. The OLS estimate (-0.63) sits on the INELASTIC side of that line, with a tight interval — so a competent classical analyst handed an endogenous

panel would confidently HOLD PRICES FLEET-WIDE. 2SLS (-1.66) puts them back on the elastic side. Endogeneity does not shade the number; it reverses the call.

What this 90% confidence interval does NOT say: that there is a 90% probability the true effect lies inside it. It is a property of the *procedure* — 90% of intervals built this way would cover the truth across repeated samples. This interval either contains the truth or it does not. To get a probability *about the effect itself* — the thing a go/no-go rule like $P(\text{lift} > \text{cost}) \geq 0.9$ actually needs — you need a posterior. That is what the Bayesian section adds.

Read-out — the classical answer, in business terms. On the clean panel (A1 holding), a pooled log-log regression with region intercepts and the §3 controls says fleet demand is **elastic**: a 1% price cut buys appreciably more than 1% of extra volume (the number is printed above), the planted truth sits inside the region-clustered 90% interval, and the whole interval lies below -1 — so at the *fleet* level, the classical arm alone is enough to say “we are leaving money on the table with a single national price.” On the endogenous panel, the same competent analyst — same controls, same clustering, no mistakes — is handed an elasticity on the **wrong side of the decision line** with a tight interval around it, and 2SLS with a cost-shifter instrument (F printed above, comfortably past the Stock–Yogo bar) hands the causal slope back at the cost of a much wider interval. That asymmetry is the whole IV bargain, and §8 will price it properly.

Notice what has **not** happened yet. No likelihood was written down, no prior was chosen, no sampler ran. The causal work was done by the *identification* — A1–A4, and an instrument when A1 fails — not by the machinery. **This is the honest baseline against which the Bayesian section must justify itself**, and 5x will hold it to that. Notice, too, exactly where the classical arm ran out of road: it gave us **one** elasticity, and the business needs **twelve**.

The guardrail — what that confidence interval does *not* say (printed above, and printed in the same words in every notebook of this cookbook): it is **not** a 90% probability that the true elasticity lies inside it. A confidence interval is a property of the *procedure* — 90% of intervals built this way would cover the truth across repeated samples — and this particular interval either covers it or it does not. Here is exactly where that bites, and it is not pedantry. §7’s pricing rule is

$$\text{set the optimal price in region } r \iff \mathbb{P}(\beta_r < -1) \geq 0.9,$$

a probability **about the elasticity itself**, evaluated region by region. The classical apparatus can report “region r ’s **one-sided 90%** confidence bound clears -1” — and 5x scores it on exactly that, at exactly this 0.9, so the two arms meet at the same bar. But *matching the level is not emitting the number*. That bound is a statement about the **procedure** — 90% of bounds built this way sit on the right side of the truth in repeated sampling — and it is emphatically **not** a 0.9 probability that region r is elastic. There is nothing to plug into a rule of the form “act when the probability exceeds x ”, and nothing to hand a CMO who asks “*how sure are we about region 5, specifically?*” That quantity does not exist in the frequentist vocabulary; it exists in a posterior. That, and not a better point estimate, is what §4 is for — and 5x will grade whether it delivers.

4 · Estimate — three pooling regimes, side by side

To make the bias–variance trade-off concrete we estimate each region’s elasticity **three ways**:

- **No pooling** — a separate regression per region. Unbiased, but a data-poor region’s estimate is pure noise. This is the “trust each region’s own numbers” extreme.
- **Complete pooling** — one elasticity for the whole fleet. Rock-steady, but pretends every region is identical, so it’s biased for any region that genuinely differs. The “everyone’s the same” extreme.
- **Partial pooling** — the hierarchical Bayesian middle ground: each β_r is nudged toward the fleet mean by an amount that depends on how much data the region has. Data-rich regions barely move; data-poor regions borrow heavily. This is the one that gets *both* the overall level and the spread right.

The model below reads directly as the maths from Step 3: `mu_beta` is the fleet-mean elasticity μ_β , `tau_beta` is how much regions differ τ_β , and `beta_r` are the per-region elasticities drawn from $\mathcal{N}(\mu_\beta, \tau_\beta^2)$. `alpha_r` are region baselines and `gamma` the controls (competitor price, seasonality, trend). The printout compares the recovered fleet mean to the planted truth.

The fitted model, in symbols (observation i in region $r[i]$; z_i the standardized controls — competitor price, season, trend):

$$\begin{aligned} \log Q_i &\sim \mathcal{N}(\alpha_{r[i]} + \beta_{r[i]} \log P_i + \gamma^\top z_i, \sigma^2) \\ \beta_r &\sim \mathcal{N}(\mu_\beta, \tau_\beta^2), \quad \alpha_r \sim \mathcal{N}(0, 5^2), \quad \gamma \sim \mathcal{N}(0, 2^2) \\ \mu_\beta &\sim \mathcal{N}(-1, 1^2), \quad \tau_\beta \sim \text{HalfNormal}(1), \quad \sigma \sim \text{HalfNormal}(1). \end{aligned}$$

The middle line’s $\beta_r \sim \mathcal{N}(\mu_\beta, \tau_\beta^2)$ **is** the partial pooling: τ_β (how much regions truly differ) is *learned from the data*, and each region’s posterior elasticity is a precision-weighted compromise between its own weeks and the fleet mean — thin regions lean on the fleet, data-rich regions stand on their own.

Why these priors. $\mu_\beta \sim \mathcal{N}(-1, 1)$ puts ~95% of its mass on $(-3, +1)$: it spans everything from mildly inelastic demand to $\beta \approx -3$ (more price-sensitive than almost any retail elasticity ever measured) while gently centering on the economically plausible region — and it is weak enough that the panel dominates it easily. $\tau_\beta \sim \text{HalfNormal}(1)$ lets the data choose the **pooling regime itself**: $\tau_\beta \rightarrow 0$ collapses to complete pooling (all regions identical), large τ_β approaches no pooling (every region on its own) — the two baselines above are the *limiting cases* of the model we actually fit, and where we land between them is *learned*, not assumed. $\alpha_r \sim \mathcal{N}(0, 5)$ is diffuse for log-scale intercepts, $\gamma \sim \mathcal{N}(0, 2)$ is weakly informative for standardized controls, and $\sigma \sim \text{HalfNormal}(1)$ is generous for log-demand residual noise (the DGP’s is 0.08).

A practitioner’s note on parametrization. We fit the *centered* form $\beta_r \sim \mathcal{N}(\mu_\beta, \tau_\beta^2)$, and here it samples cleanly (check the divergence count in the convergence report below). But centered hierarchies are *the* canonical NUTS pitfall: with few groups or weak data the joint posterior of (τ_β, β_r) develops a **funnel** — as $\tau_\beta \rightarrow 0$ the β_r are squeezed into an ever-narrower band the sampler cannot enter. The standard fix is the **non-centered** form $\beta_r = \mu_\beta + \tau_\beta \tilde{\beta}_r$ with $\tilde{\beta}_r \sim \mathcal{N}(0, 1)$, which decorrelates the geometry. If you port this notebook to a panel with 4 regions and 12 weeks, switch to non-centered first.

Like-for-like baselines. The no-pool and complete-pool baselines are fit with the **same standardized controls** Z (competitor price, season, trend) as the hierarchical model — per-region OLS and pooled OLS with region intercepts, respectively — so the MAE comparison in step 6 isolates the **pooling choice** and nothing else. (An earlier draft used bivariate $\log Q$ -on- $\log P$ fits

for the baselines, which conflated the pooling choice with the control choice; clean comparisons are four lines of code, so we make them.)

```
hierarchical model convergence: max r-hat 1.000 - min ESS 675 - divergences 0
fleet-mean elasticity mu_beta -1.36 (true -1.42) * between-region spread
tau_beta 0.30 (true 0.31)
```

Reading the convergence line. Those three numbers are the sampler’s health check — whether the MCMC sampler (the algorithm drawing from the posterior) actually converged. **R-hat** compares the variance *within* each chain to the variance *across* chains; 1.00 is perfect and ≤ 1.01 is **the usual pass bar** (above it the chains disagree and the numbers aren’t safe to read). **ESS** (effective sample size) is how many *independent* draws our autocorrelated chains are worth — a few hundred is ample for a posterior mean or a 90% interval. **Divergences** are steps where the sampler broke down and silently distrusts that region — for this centered hierarchy they are the funnel warning flagged above, and **you want 0**. All three pass here, so the estimates below are safe to read. (Under the FAST teaching profile the chains are short, so R-hat can drift to ~ 1.02 and PyMC may print a “problems during sampling” notice — a benign artifact of the small draw count that clears in a FULL run.)

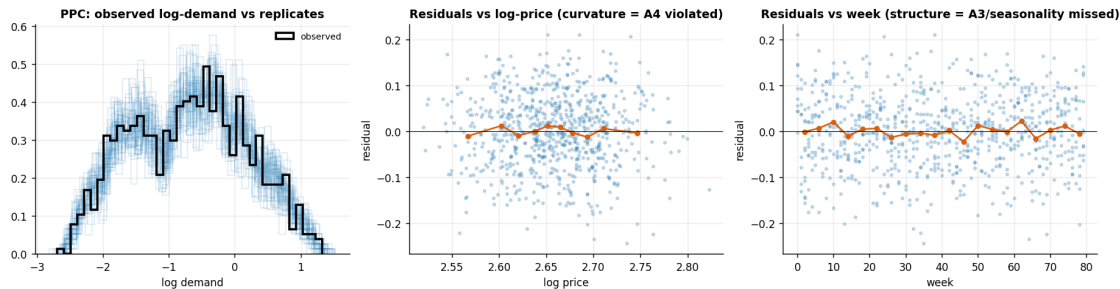
5 · Check the model *before* grading it — posterior predictive & functional form

Discipline first: before we congratulate the model on recovering the planted truth (a luxury a real analyst never has), we run the checks a real analyst *can* run.

- **Posterior predictive check (PPC).** Simulate replicated datasets $\log Q^{\text{rep}}$ from the fitted model and overlay their distribution on the observed $\log Q$. If the cloud of replicates fails to cover the data — wrong location, spread, or shape — the model is structurally wrong somewhere.
- **Residuals against log-price and against week.** Curvature in residuals-vs- $\log P$ is exactly what a violated constant-elasticity assumption (A4) looks like; leftover structure in residuals-vs-week means missed seasonality or dynamics (A3).
- **A curvature term as a formal probe of A4.** We add $(\log P - \overline{\log P})^2$ to a pooled OLS with the same controls and check whether its coefficient is distinguishable from zero.

And the sentence that matters most: a clean PPC certifies **fit, not causality**. The endogenous model of step 8 — wrong by nearly a full elasticity point — would sail through this same PPC, because confounding is invisible in the distribution of the outcome alone. PPC failures tell you the model is wrong; PPC successes do **not** tell you it is right about the causal question. That was assumption A1, and A1 is untestable. Keep the two firmly apart.

A4 curvature probe - coefficient on $(\log P - \text{mean})^2$: -0.9 (SE 0.8, $t = -1.0$) -> consistent with zero -> constant elasticity survives in-sample
note the wide SE: with only $\pm 5\%$ price wiggle this test has little power far from current prices -
another reason to treat any P^* outside the observed range as extrapolation, not measurement.
PPC replicate mean -0.70 vs observed -0.70; replicate SD 0.88 vs observed 0.88



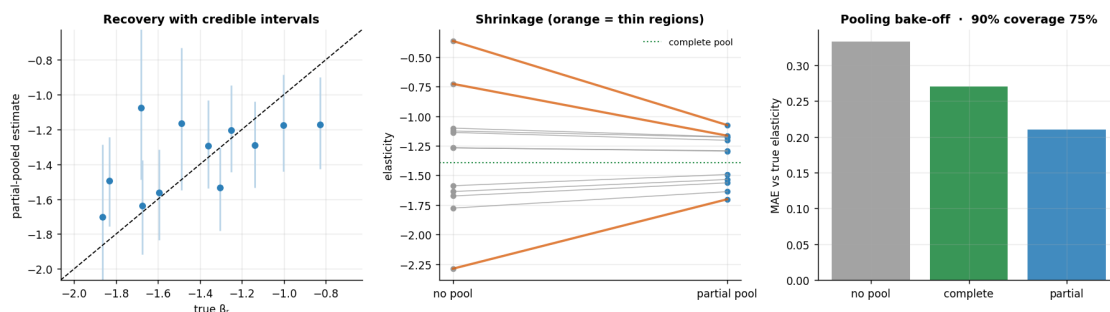
How to read the three panels. *Left* — the black observed log-demand density sits inside the blue cloud of posterior replicates: the model reproduces the location, spread and shape of the outcome. *Middle* — residuals vs log-price show no systematic bend: the binned means (orange) hug the zero line across the observed price range, so constant elasticity is consistent with the data *where we have data*. *Right* — residuals vs week show no leftover seasonal wave or drift; the sine/trend controls are doing their job. The curvature probe printed under the figure makes the middle panel formal — and its wide standard error is itself informative: with only $\pm 5\%$ of price variation, in-sample data can never certify the demand shape far from current prices. **None of this validates A1 (exogeneity):** the step-8 model passes every one of these checks while being badly wrong about the causal slope.

6 · Validate — recovery, shrinkage, and calibration

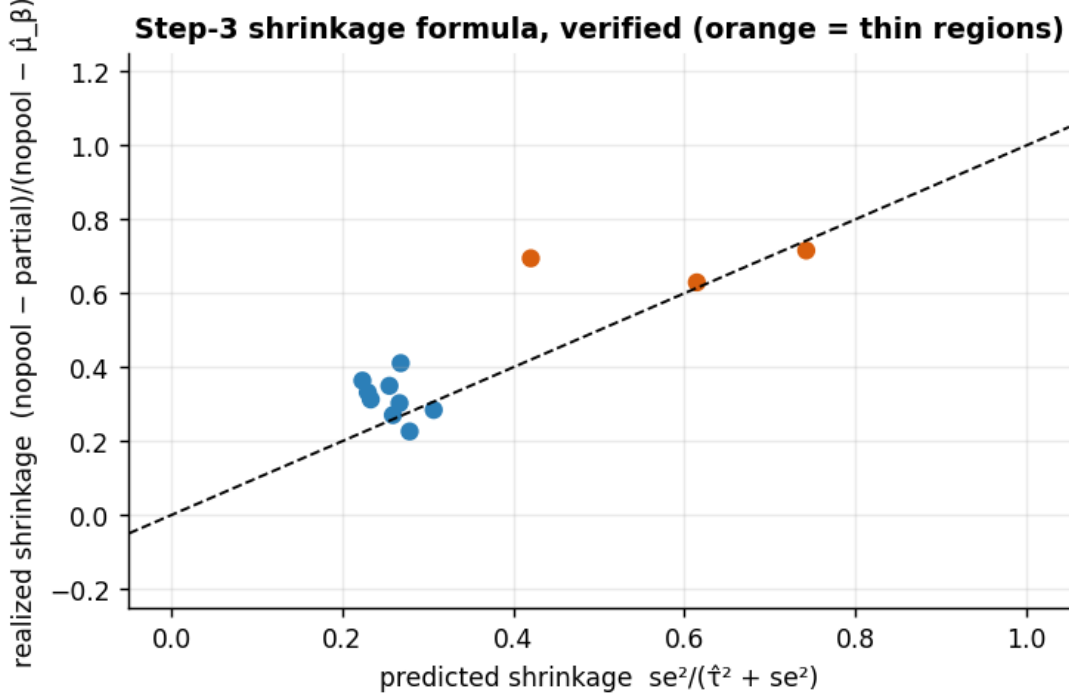
Estimated vs true elasticity should track the 45° line; the thin regions' point estimates get pulled toward the fleet mean (that's pooling protecting us), and the credible intervals are wider where data is thin. We also check that the 90% intervals actually **cover** the true elasticities (calibration).

MAE - no pool 0.33, complete 0.27, partial 0.21 * coverage 75%

coverage split - thin regions (0-2): 2/3 covered * rich regions: 7/9 covered
with only 12 regions, a perfectly calibrated 90% interval covers 9 or fewer with probability 0.11 -> within Monte-Carlo noise of calibrated



thin regions - predicted shrinkage [0.74 0.61 0.42], realized [0.72 0.63 0.69]
rich regions - predicted 0.26 on average, realized 0.32 (0 region(s) hidden: no-pool slope already ~ fleet mean, so the ratio is ill-defined)



How to read those panels. *Left* — partial-pooled estimates track the true elasticities along the 45° line, and the error bars are visibly wider for the thin regions (less data → more honest uncertainty). *Middle* — the shrinkage in action: lines connect each region’s noisy no-pool estimate (left) to its partial-pooled estimate (right); the **thin regions (orange)** get yanked hardest toward the fleet, while data-rich regions barely budge. That selective pull is exactly the protection partial pooling buys. *Right* — the bottom line: mean absolute error against the truth is **lowest for partial pooling**, beating both extremes *even now that the baselines get the same controls* — no-pooling is noisy, complete-pooling is biased, and the remaining gap is the pooling choice alone. **This bar chart is one panel**, though, and one panel is an anecdote: 5x re-runs the whole bake-off on 25 fresh ones (with a full hierarchical refit each time) and reports the win *and* its spread. Read that before quoting a margin from this figure.

Is the below-nominal coverage a bug? Don’t wave it away — check it, twice; both checks are printed under the figure. The **split** locates any misses: shrinkage intervals are calibrated for the *ensemble*, not for each pre-shrinkage slope, so intervals run too narrow precisely for regions whose true elasticity sits far from the fleet mean and for the hard-shrunk thin ones. The **binomial line** asks whether the shortfall is even signal: with 12 regions, a perfectly calibrated 90% procedure still misses a couple regularly, and the printed probability says on which side of that line this run falls. When it flags a *real* shortfall, that is the known price of shrinkage — slightly overconfident **per-region** intervals bought in exchange for the best **point** estimates (lowest MAE, right panel) — worth knowing before quoting an interval to a region manager, and one more reason the decision rules in step 7 lean on $P(\beta_r < -1)$ across the whole posterior rather than on interval endpoints. The follow-up scatter then closes the loop on step 3’s shrinkage formula: the predicted pull $se_r^2/(\hat{\tau}_\beta^2 + se_r^2)$ — which is exactly the step-3 factor rewritten, since se_r stands in for $\sqrt{\sigma^2/n_r}$ and $1 - \tau^2/(\tau^2 + se^2) = se^2/(\tau^2 + se^2)$ — broadly tracks the pull the posterior actually applied (near the 45° line — the

approximate se-based formula is roughest for the smallest-n region), thin regions (orange) toward the heavy-shrinkage corner — the formula we asked you to trust in step 3 is the formula the sampler obeyed.

5x · Point estimate vs posterior — what the Bayesian layer actually bought

Step 0 gave the classical arm both of its options for a per-region elasticity: **complete pooling** (one number, quoted twelve times) and **no pooling** (twelve independent regressions, three of them on 16 weeks). §4 gave the Bayesian arm the third: **partial pooling**. All three target the identical estimand under the identical assumptions. Below they meet, region by region, and are graded against the planted truth — and then against the only question the business actually asked.

The decision rules put a fair, *unhelpful-to-us* version of the classical arm on the field. §7's rule is $\mathbb{P}(\beta_r < -1) > 0.9$ — a **one-sided 90%** requirement — so the classical arms are scored on the **one-sided 90% confidence bound**, $\hat{\beta}_r + t_{0.90, df} \hat{se}_r < -1$: the same 90%, on the same side. (The tempting alternative — “the two-sided 90% CI lies below -1 ” — is secretly a one-sided **95%** test, a stricter bar, and scoring the classical arm there would win the argument with a threshold instead of an estimator. The cell prints both, so you can see exactly what that costs.) If the posterior's advantage is real, it has to show up against the *matched* bar.

complete pooling says -1.39 for every region below [90% CI -1.54, -1.24].

region	n	true	no-pool	no-pool 90% CI	partial	partial 90% CrI	
P(beta<-1)							
region_00	16	-1.68	-0.36	[-1.31, +0.58]	-1.07	[-1.49, -0.56]	0.64
region_01	16	-1.86	-2.29	[-2.65, -1.93]	-1.70	[-2.23, -1.29]	1.00
region_02	16	-1.49	-0.73	[-1.11, -0.34]	-1.16	[-1.55, -0.73]	0.75
region_03	80	-1.25	-1.14	[-1.46, -0.82]	-1.20	[-1.45, -0.95]	0.91
region_04	80	-1.00	-1.12	[-1.51, -0.74]	-1.18	[-1.44, -0.89]	0.84
region_05	80	-1.36	-1.27	[-1.54, -1.00]	-1.29	[-1.54, -1.03]	0.97
region_06	80	-1.59	-1.67	[-1.93, -1.42]	-1.56	[-1.84, -1.32]	1.00
region_07	80	-1.67	-1.78	[-2.09, -1.46]	-1.64	[-1.92, -1.37]	1.00
region_08	80	-1.14	-1.27	[-1.56, -0.98]	-1.29	[-1.53, -1.04]	0.97
region_09	80	-0.83	-1.10	[-1.46, -0.74]	-1.17	[-1.42, -0.90]	0.86
region_10	80	-1.30	-1.64	[-1.90, -1.37]	-1.53	[-1.78, -1.30]	1.00
region_11	80	-1.83	-1.59	[-1.86, -1.32]	-1.49	[-1.76, -1.24]	1.00

ACCURACY (MAE vs the twelve planted betas) - reproducing section 6's bake-off with the cl toolkit:

```
no pooling      0.333
complete pooling 0.270
partial pooling  0.210
```

-> best: partial pooling; on THIS panel partial pooling beats the better classical arm by +22%.

ONE panel, one seed, twelve regions, no interval - exactly the reasoning Step 0 refused.

The next cell re-runs all three arms on fresh panels; quote THAT number, not this one.

the no-pooling arm's worst region is region_00 (16 weeks): it estimates -0.36

against a true -1.68;

the hierarchical model, borrowing from the fleet, puts it at -1.07.

interval width (mean): no-pool 0.74 vs posterior 0.61; on the three THIN regions 1.13 vs 0.90.

THE DECISION - of the 11 genuinely elastic regions ($\beta < -1$), how many does each

arm act on, and does it wrongly act on the 1 that is not?

arm	acts on	correct	WRONG	missed
classical * complete pooling (1-sided 90% bound)		12	11	1
0				
classical * no pooling (1-sided 90% bound)	7	7	0	4
Bayesian * partial pooling ($P > 0.9$)	8	8	0	3

THE BAR IS NOT THE ESTIMATOR. Score the same no-pooling fits on 'the TWO-SIDED 90% CI lies below

-1' instead - the natural-looking choice, and a one-sided 95% test - and the arm acts on only 5,

not 7: region_05, region_08 drop out on the threshold alone (their 1-sided 90% bounds are -1.06, -1.04, just past -1).

Held at the SAME 90%, the honest scoreline is classical 7 vs Bayesian 8 - a gap of 1 region(s), not the 3 the mismatched bar would have sold you.

...and the column that does not exist. The two classical rows are now scored at the SAME 90%

as the posterior - the counts are like-for-like. But matching the *level* does not make them the

same *quantity*. 'The one-sided 90% bound clears -1' is a property of the PROCEDURE (90% of such

bounds cover the truth in repeated sampling); it is not a probability that THIS region is elastic.

Only the last column emits a number in [0,1] about the elasticity itself - the number section 7's

rule, and its 0.5 'controlled test' tier, are actually written in.

MAE vs the planted betas over 25 FRESH panels (each one a full hierarchical refit) - mean +/- sd:

no pooling 0.190 +/- 0.054 (this panel, seed 5: 0.333)

complete pooling 0.292 +/- 0.065 (this panel, seed 5: 0.270)

partial pooling 0.159 +/- 0.036 (this panel, seed 5: 0.210)

-> partial pooling beats the better classical arm on 19/25 panels, by +9% on average

(sd 24 pp; worst panel -63%, best +43%).

-> the single-seed headline above (+22%) sits at the 68th percentile of that spread - so quote the multi-panel line, never the one-panel one.

-> head-to-head: partial pooling beats no pooling on 19/25 panels, complete pooling on 25/25.

-> and WHICH classical arm is the one to beat is itself a one-seed artefact:

on this panel it is

complete pooling (0.270), but averaged over 25 panels it is no pooling (0.190 vs 0.292).

(convergence across all 25 refits: max r-hat 1.010, 25 divergences in total.)

The honest verdict — what the posterior bought, and what it did not.

1 · On the fleet number, they agree — say it plainly. Step 0’s pooled OLS and §4’s posterior μ_β land on essentially the same fleet elasticity, both within a hair of the planted truth. That is not a disappointment; it is the lesson. **The causal work was done by the identification, not by the machinery.** With 768 region-weeks and weak priors, a Bayesian hierarchy is not going to move a well-identified pooled slope, and it does not. If you came here hoping “going Bayesian” would improve the headline number, this is the correction.

2 · On the *twelve* numbers, partial pooling wins — but read the 25-panel line, not this panel’s. The single-panel MAE block above is seductive, and it is *one draw*. Step 0 spent ten lines refusing to read bias off one panel (“one sample cannot separate bias from luck”), and the headline number does not get an exemption: the cell above re-runs **all three arms on 25 fresh panels**, each with a full hierarchical refit at the same sampler settings. The win is real, and it is smaller and noisier than this panel advertised:

- Partial pooling has the **lowest mean MAE** of the three (≈ 0.16 , against ≈ 0.19 for no pooling and ≈ 0.29 for complete pooling) *and* the **smallest seed-to-seed spread**. It is the most accurate arm and the most stable one — the two things you want from an estimator you will re-run every quarter.
- It beats the better classical arm on **19 of 25 panels**, and that comparator is deliberately *unhelpful to us*: it is the **oracle-best** classical arm, re-chosen panel by panel against the planted truth an analyst would never have. Partial pooling has to beat whichever extreme happened to get lucky.
- The average margin is **+9%, not the +22% this panel prints** (sd ≈ 24 pp; on its worst panel partial pooling is *worse* by 63%). Seed 5 sits around the 68th percentile of that spread. **The estimator wins; the headline was flattered by the draw.** Six panels in twenty-five, it loses.
- And the 25 panels overturn something else this table implies. *Which classical arm is the one to beat is itself a one-seed artefact.* Here complete pooling (0.270) looks better than no pooling (0.333) — but only because region_00’s sixteen weeks produced a spectacular miss (-0.36 against a true -1.68) that drags the no-pooling average up single-handed. Averaged over panels the ranking **reverses**: no pooling 0.190, complete pooling 0.292. With 80 weeks in nine of twelve regions, “trust each region’s own data” is usually the better classical bet, and complete pooling’s identical-for-everyone bias is the steadier drag. The blow-up in the table is the *anecdote*; the mean is the *fact*.

So the claim survives contact with 25 panels, and it lands quieter than one panel would let you say it: partial pooling interpolates between the two extremes with a weight the data chooses (τ_β), and that interpolation is worth roughly **9% of MAE on average and a visibly tighter spread** — not 22%. **This is still the one place in the notebook where the Bayesian layer moves the estimate, and it moves it because the classical toolbox has no third option to offer** — not because its arithmetic is better. (A frequentist mixed model would get close; it is, after all, the same likelihood. The point stands that plain OLS cannot.)

3 · On the decision, the classical arm *can* play — once you stop rigging the bar. §7’s

rule is $\mathbb{P}(\beta_r < -1) \geq 0.9$: a **one-sided 90%** requirement. The tempting way to score the classical arm is “*its 90% CI lies entirely below -1*” — and it is the wrong way. A two-sided 90% interval puts 5% in each tail, so that test is secretly a one-sided **95%** bar: strictly stricter than what the posterior is being asked to clear. Score the arms that way and the posterior “wins” **8 regions to 5** — and three of those regions were handed over by the *threshold*, not won by the *estimator*. Scored where it belongs, on the matched one-sided 90% bound $\hat{\beta}_r + t_{0.90,df} \hat{se}_r < -1$ (same 90%, same side), no pooling acts on **7** regions against the posterior’s **8**, and *neither arm ever prices the one truly inelastic region*. `region_05` and `region_08` — bounds -1.06 and -1.04 — were never “missed by the classical estimator”; they were missed by a mismatched bar. **The honest gap is one region**. It still points the same way, which is why the notebook’s conclusion survives; it just stops being inflated.

Where the classical arm *does* fail on this decision, it fails expensively, and fair scoring does not save it: **complete pooling’s bound also clears -1, so it certifies all twelve regions as elastic** — including the one the DGP planted as genuinely *inelastic*, which would get a price rise it cannot support. That is a **wrong** action, not a missed one, and it is the only wrong action anywhere in the table. No pooling, scored fairly, leaves 4 of the 11 elastic regions on the table; the posterior leaves 3. On counts, that is a rounding error — and it is the right moment to stop pretending the win is about counts.

And the dial exists — it simply does not turn the knob the CMO needs. It would be convenient to say the classical arm has no threshold to tune. It has one: α . An analyst can build an 80% bound, or test at any level she likes, and the roster re-sorts — exactly as the posterior’s roster re-sorts when the action bar moves from 0.9 to 0.8. The difference is *what the two dials control*. Turning α tunes the **long-run error rate of the procedure**: at $\alpha = 0.1$, one in ten bounds built this way lands on the wrong side of the truth, **across hypothetical repetitions of the whole study**. It cannot be set to a statement of risk appetite about **region_05 in particular**, because “the probability `region_05` is elastic” is not a quantity the procedure emits — `region_05`’s bound either clears -1 or it does not, and which of those happened is not something the interval can tell you. The posterior’s dial is the other thing entirely: $\mathbb{P}(\beta_{05} < -1) = 0.97$ is a number **about that region**, in the only sample we have, and moving the bar to 0.8 changes *which regions we are willing to be wrong about, and how often*. Matching the **level** of the two bars — which 5x does, honestly — does not make them the same **quantity**. That is the durable difference, and it is the one that survives the corrected scoreline.

That is also what pays for §7’s third tier. Because each region arrives with a number in $[0, 1]$, the pricing rule can **route** it: act above 0.9, run a **controlled test** between 0.5 and 0.9, hold below. A confidence bound has no “how sure are we” to report — it is in or out — so the classical arm can only build a two-tier policy. The posterior’s three misses are not a failure; they are a **choice**, made against a stated probability, and the regions it holds back are the ones §7 sends to a price test rather than abandons.

What Bayes did NOT buy here. Not the fleet elasticity (Step 0 had it, correctly, with an honest clustered interval). Not a better standard error — clustering already told us the pooled model was misspecified, which was the *same message* the hierarchy acts on, just delivered without a way to respond to it. Not protection from endogeneity: the §4 posterior is exactly as biased as the OLS slope when A1 fails (§8 proves it), and the fix is an instrument, which Step 0 fit with no prior in sight. Not, on a matched bar, a landslide on the decision count: that margin is **one region**, and the 8-vs-5 scoreline a mismatched threshold would have handed us was not a finding, it was an accounting error. What it *did* buy is the **third pooling regime** — worth ~9% of MAE on

average, and a materially tighter spread across panels — and the **probability statement**: twelve honest per-region elasticities instead of one, and a $\mathbb{P}(\beta_r < -1)$ per region that a pricing rule can actually be written in. On the euros, the edge over a fairly-scored classical arm is narrow. On the **vocabulary**, the gap is total — and it is the vocabulary, not the euros, that lets §7 run a three-tier *set / test / hold* policy instead of a coin flip. That is the entire win, honestly sized. For this business problem, it is still the one worth having.

7 · Decide, in euros — profit-maximising price per region

The decision-maker’s question is not “what is β ?” but “**what price do I set?**” For constant-elasticity demand $Q(P) = AP^\beta$ and constant marginal cost c , weekly profit is

$$\pi(P) = (P - c)Q(P) = (P - c)AP^\beta.$$

Differentiate and set to zero:

$$\frac{d\pi}{dP} = AP^\beta + (P - c)\beta AP^{\beta-1} = AP^{\beta-1}[(\beta + 1)P - \beta c] = 0 \implies P^* = \frac{\beta}{\beta + 1}c,$$

valid **only when** $\beta < -1$ (demand elastic). The bracket $(\beta + 1)P - \beta c$ then starts positive at small P and decreases through zero — profit rises, peaks at P^* , falls: a genuine maximum (that sign change *is* the second-order condition). For $-1 \leq \beta < 0$ the bracket stays positive for every $P > 0$, so the model says “raise price without limit” — which really means *the constant-elasticity model has left its domain of validity*, not that infinite prices are optimal.

Rearranged, P^* is the classic **Lerner index** (markup) rule:

$$\frac{P^* - c}{P^*} = -\frac{1}{\beta}$$

— *optimal markup over cost equals the inverse elasticity*, the oldest one-liner in pricing: $\beta = -2$ prices at twice cost, $\beta = -1.5$ at three times cost. And note the **blow-up as** $\beta \rightarrow -1^-$: the denominator $\beta + 1 \rightarrow 0^-$ and $P^* \rightarrow +\infty$. Posterior draws near -1 therefore produce wild optimal prices — the statistical reason (exploited below) why regions whose elasticity posterior straddles -1 get routed to a **controlled test**, not a price move.

We propagate the **elasticity posterior** draw-by-draw into a **price posterior** and a **profit-uplift posterior** per region, and — the honest part — flag regions where we’re not confident demand is even elastic ($\beta < -1$) for that test rather than acting on a shaky slope.

region	elasticity	P(elastic)	P* €	P* 90% CI	uplift %	action
region_00	-1.07	0.64	-	-	34	controlled test
region_01	-1.70	1.00	19.9	[14.5, 35.6]	13	set price
region_02	-1.16	0.75	-	-	35	controlled test
region_03	-1.20	0.91	43.7	[25.7, 220.7]	46	set price
region_04	-1.18	0.84	-	-	43	controlled test
region_05	-1.29	0.97	34.2	[22.8, 111.5]	36	set price
region_06	-1.56	1.00	22.4	[17.6, 33.3]	16	set price

region_07	-1.64	1.00	20.6	[16.7, 29.3]	12	set price
region_08	-1.29	0.97	34.6	[22.9, 107.8]	37	set price
region_09	-1.17	0.86	-	-	45	controlled test
region_10	-1.53	1.00	23.1	[18.2, 34.2]	17	set price
region_11	-1.49	1.00	24.5	[18.6, 40.7]	20	set price

Across the 8 confidently-elastic regions, moving to optimal pricing lifts profit by +24%

[90% +15%, +35%] on average - the pricing change's value, scale-free and with uncertainty carried through (assumes constant elasticity out to P^* ; see caveats).

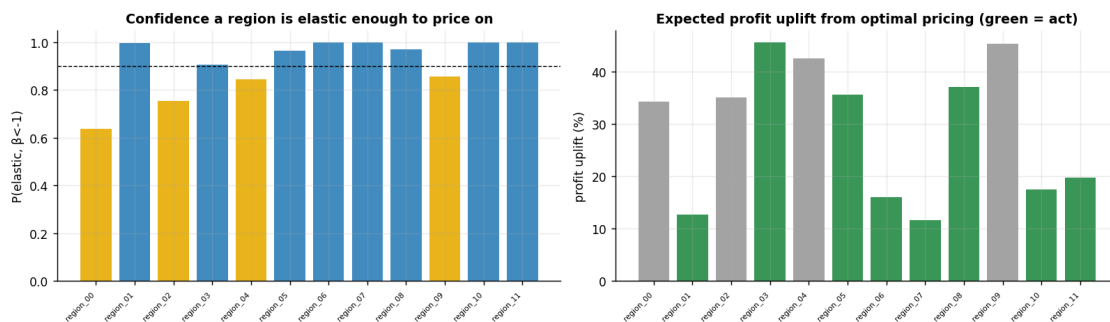
point-plug illustration - weakest region region_00: $P(\text{elastic}) = 0.64$, mean beta = -1.07;

plugging that mean into $P^* = c\beta/(\beta+1)$ gives £116, vs £22 for the most confident region (region_06).

its expected uplift +34% is diluted by the 36% of draws pinned at 0%;

conditional on truly

being elastic it is +54% - a big, *unreliable* number, which is exactly why it goes to a test.

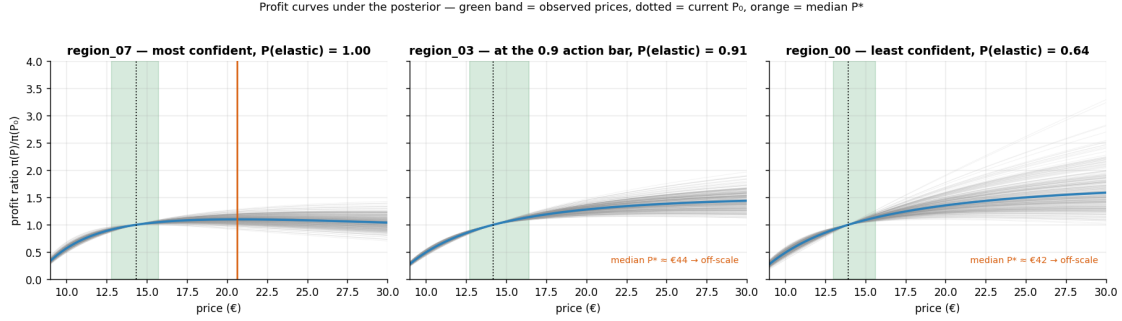


Green band = prices ever charged; the model was only ever tested there (step 5).

Both P_0 (dotted) and the

median P^* (orange) matter less than the SHAPE: a flat top means a mis-estimated beta costs little profit;

curves that fan out and keep rising off the right edge mean the 'optimum' is an extrapolated guess.



How to read the decision table and the two panels. *Left* — each bar is $P(\beta < -1)$, our posterior confidence that a region’s demand is **elastic** enough to price on: **blue** clears the 0.9 line (*set price*), **gold** sits in 0.5–0.9 (*controlled test*), **orange** is below 0.5 (*hold/test*). *Right* — the expected **profit uplift** from moving to the optimal price, **green** only where we are confident enough to act. The fleet number underneath averages the uplift across the confidently-elastic regions and carries the posterior through to a 90% interval. The **P*** columns are the number the CMO actually signs: the posterior-median optimal price with its 90% credible interval — printed *only* for regions confident enough to act on (“—” elsewhere: publishing a price for a region we just routed to a test would invite exactly the mistake this notebook is built to prevent).

One subtlety worth saying out loud — the kind a sharp audience member will probe. The “profit uplift %” is a **posterior mean over every draw**, with the non-elastic draws ($\beta \geq -1$, where raising price doesn’t pay) contributing **0%**. So it is an *expected* uplift already discounted by our uncertainty — not “the uplift once we act”. For the confident **set price** rows ($P(\text{elastic}) \approx 1$) the two coincide; for the weakest region — see the *point-plug illustration* printed under the table — the expected uplift is diluted by the slice of draws worth 0%, and the uplift *conditional on the region truly being elastic* (also printed) is far larger and far shakier.

This is also why you **cannot reproduce the uplift column by plugging the row’s own elasticity into the formula**. The closed form $P^* = c\beta/(\beta + 1)$ blows up as $\beta \rightarrow -1^-$: the printed illustration plugs the weakest region’s *mean* elasticity into the formula and gets an absurd price, a world away from the sane optima of the well-elastic regions. So the per-draw uplift has a heavy right tail and its **mean is not the formula evaluated at the mean elasticity (Jensen’s inequality)** — for a curved function the average of the outputs is not the function of the average input; here P^* bends so sharply near $\beta = -1$ that the mean of the per-draw uplifts sits well above the uplift computed at the mean β). We deliberately propagate the *whole* posterior rather than point-plug a single slope — and that very instability near $\beta = -1$ is the statistical reason those low-confidence regions are routed to a **controlled test**, not to a price move we might regret.

The profit curves make the same point visually. In the *most confident* panel the grey posterior curves agree on a gentle hill with a **flat top**: even a sizeable error in β moves profit little near P^* , so acting is low-risk. In the *least confident* panel the curves fan out and many keep rising off the right edge — the model is begging for a price far outside the green band of prices we have ever charged, on a functional form (A4) we could only test *inside* that band; statistical risk and extrapolation risk stack, so it goes to a test. The *middle* panel sits near the 0.9 action bar and is the boundary case the thresholds are built around: most draws agree on a gentle hill, but a visible minority still explode off to the right — whether it clears the bar is decided by the printed

$P(\beta < -1)$, and standing this close to the threshold is itself an argument for measuring before committing. And notice one sobering detail: even for confident regions P^* typically sits *above* the observed price band — the recommendation itself is an extrapolation. That is why a staged rollout (move a minority of stores first, watch a few weeks) beats a fleet-wide overnight change even where the posterior is sure.

The cost assumption — the other number in the formula

$P^* = c\beta/(\beta + 1)$ is **linear in marginal cost c** : a $\pm 25\%$ error in c moves *every* recommended price by $\pm 25\%$, no posterior required. Cost is usually handed to the analyst as a single accounting number, and it deserves the same scepticism as the elasticity — arguably more, because nothing in the data can check it. Below we sweep $c \pm 25\%$ around our €8 and recompute the whole decision layer. Two things to watch: the **action routing never changes** (it keys off $P(\beta < -1)$, which involves no c), but every P^* and every uplift number moves — so if finance restates the cost, the *set-vs-test split* survives while the price list is rewritten wholesale. Cost measurement is as decision-critical as the elasticity itself.

marginal cost €	n	set-price	median P^*	range €	fleet uplift	% uplift	90% CI
6.0		8	14.9-32.8		8	[+3%, +14%]	
6.5		8	16.2-35.5		10	[+5%, +18%]	
7.0		8	17.4-38.2		14	[+7%, +22%]	
7.5		8	18.7-41.0		19	[+11%, +28%]	
8.0		8	19.9-43.7		24	[+15%, +35%]	
8.5		8	21.1-46.4		32	[+20%, +44%]	
9.0		8	22.4-49.1		40	[+27%, +55%]	
9.5		8	23.6-51.9		51	[+36%, +68%]	
10.0		8	24.9-54.6		65	[+47%, +84%]	

reading it: the set-price roster is c -invariant by construction (column 2 never moves), and every median P^* scales exactly linearly in c - get the cost 25% wrong and every recommended price is 25% wrong, with the uplift estimate moving too. Before signing the price list, audit the cost number.

8 · Caveats — what partial pooling does *not* fix

Partial pooling handles *noise*, not *bias*. If price is endogenous to an **unobserved** demand shock, adjusting for observed Z isn't enough and the elasticity is biased **toward zero** — by step 3's OVB formula, $\hat{\beta} \rightarrow \beta + \kappa\sigma_u^2/(\kappa^2\sigma_u^2 + \sigma_v^2)$. We now do three things: **(i)** switch the hidden shock on at the demo strength $\kappa = 1.2$ and refit the full hierarchical model, checking the observed bias against the formula; **(ii)** sweep κ to map the whole bias curve — a range, not an anecdote; **(iii)** return to the **IV cure Step 0 already fit** — the same $\kappa = 1.2$ panel, the same planted cost shifter — and read what it costs. (These re-runs use the **full** panel, not the thinned subset used above — which only sharpens the fleet estimate and doesn't change the bias point being made.)

clean fleet elasticity -1.36 (true -1.42)
 endogenous fleet elasticity -0.59 (true -1.42) -> biased toward zero by +0.83
 OVB formula check - predicted bias +0.83 vs observed +0.83: the bias is

computable, not mysterious.

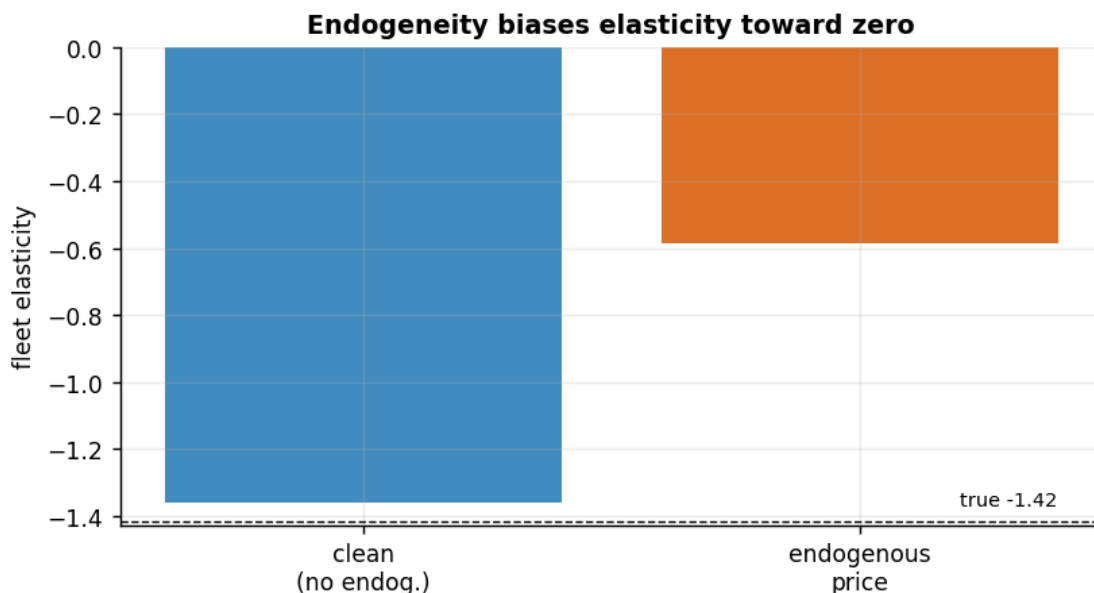
endogenous-fit convergence: max r-hat 1.000 - min ESS 5640 - divergences 0
the Bayesian hierarchy is no protection: its fleet elasticity (-0.59) sits within 0.01 of naive OLS

on the SAME raw kappa = 1.2 panel (-0.60) - partial pooling shrinks NOISE, and this is BIAS.

(Step 0's -0.63 is that same OLS on the instrument-augmented panel - the one with the cost shifter

added to log-price - so it differs by 0.03. Same story, different panel; the panel below is Step 0's.)

Fix = an instrument (a cost shifter) or a natural price experiment - Step 0 fit it; the bargain is priced below.



From a point to a curve. One κ is an anecdote. Below we sweep the endogeneity strength from 0 to 1.25 — regenerating the panel and re-estimating the fleet elasticity at each value (a controls-included pooled OLS with region intercepts stands in for the full hierarchical refit: it agrees with it at the $\kappa = 1.2$ point above and costs milliseconds instead of minutes) — and overlay the OVB *prediction* from step 3. Watch two things. (i) The estimate does not drift gently: because the exogenous price wiggle is tiny ($\sigma_\nu = 0.05$), even a faint hidden shock swamps it — the bias term $\kappa\sigma_u^2/(\kappa^2\sigma_u^2 + \sigma_\nu^2)$ peaks near $\kappa \approx \sigma_\nu$ — and the fitted “demand curve” actually slopes *upward* over a whole range of κ . (ii) The recommendation flips almost immediately: the moment the estimate crosses the $\beta = -1$ decision line, “set optimal prices in elastic regions” becomes “demand looks inelastic — hold everywhere”. Endogeneity doesn’t just shade the number; it reverses the pricing call.

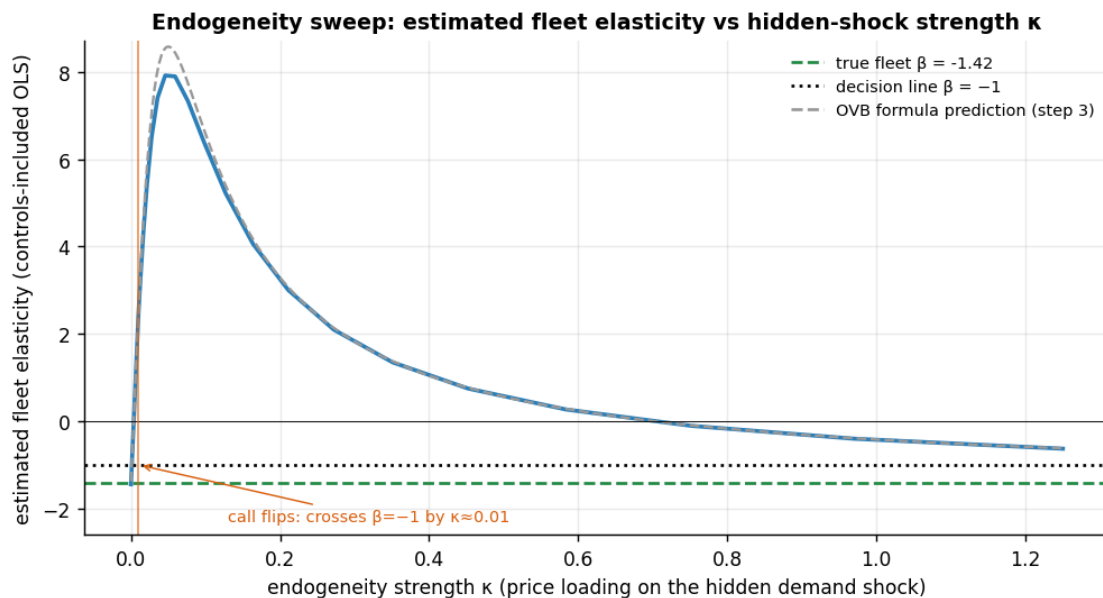
estimated fleet elasticity by kappa: kappa=0: -1.45 * kappa=0.01: +2.45 *
kappa=0.0128933: +3.31 * kappa=0.0166237: +4.31 * kappa=0.0214334: +5.41 *
kappa=0.0276347: +6.50 * kappa=0.0356302: +7.41 * kappa=0.0459391: +7.92 *

kappa=0.0592306: +7.90 * kappa=0.0763677: +7.33 * kappa=0.098463: +6.37 *
kappa=0.126951: +5.22 * kappa=0.163682: +4.06 * kappa=0.21104: +3.00 *
kappa=0.2721: +2.10 * kappa=0.350826: +1.35 * kappa=0.45233: +0.75 *
kappa=0.583202: +0.27 * kappa=0.751939: -0.11 * kappa=0.969497: -0.40 *
kappa=1.25: -0.63

the estimate is already on the wrong side of $\beta=-1$ by $\kappa \sim 0.01$: a hidden shock explaining only ~4% of residual price variance flips the pricing call fleet-wide.

the grey dashed curve is step 3's OVB formula - theory and refit agree across the whole sweep,

including the perverse region where the fitted demand curve slopes upward.



The cure, priced — the instrument from Step 0

Step 0 already did this fit, deliberately, *before* any prior appeared: the three conditions an instrument must satisfy (relevance, exclusion, independence), the 2SLS two-stage algebra, the planted cost shifter W , and `cl.iv_2sls`'s first-stage F are all up there — reread them if they have faded. Nothing is re-estimated here. What we do now is put the two classical fits side by side **with the sweep above in mind**, because only now can we see what the instrument is actually buying and what it charges for it.

first-stage $F = 27$ (>10: relevant, not weak)

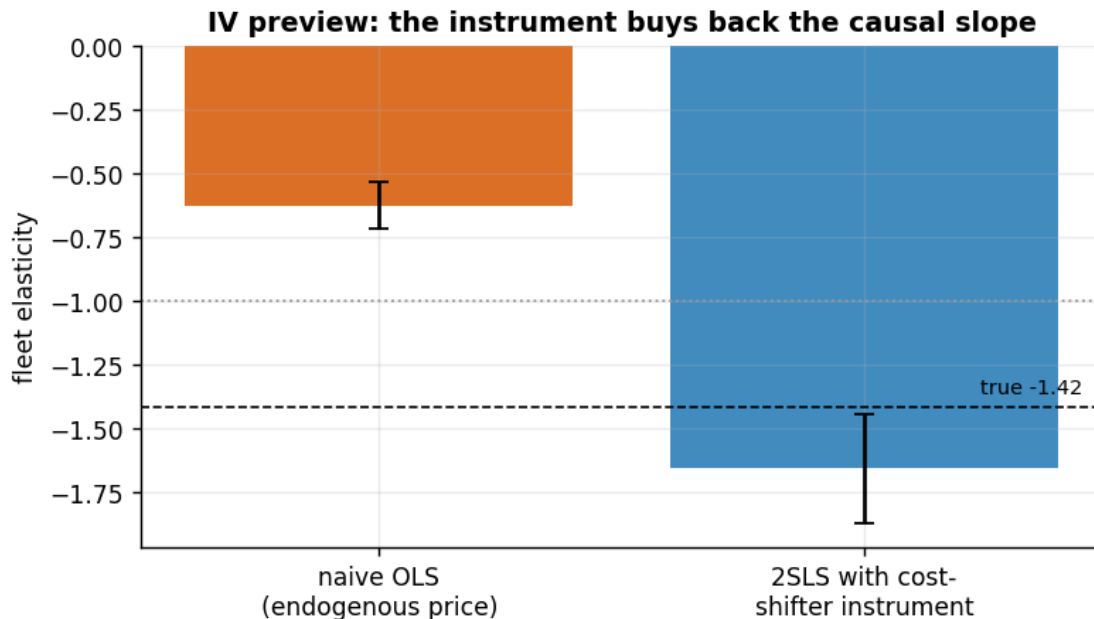
naive OLS: -0.63 ± 0.09 - a tight band 9 SEs from the truth -1.42 (precisely wrong)

2SLS: -1.66 ± 0.21 - a wide band 1.1 SEs from the truth (imprecisely right)

the cure's price, in width: the 2SLS interval is x2.3 the OLS one - it keeps only the cost-driven

slice of price variation, so the hidden-shock slice drops out of the estimate, and so does most of the data.

Full treatment - robust SEs, weak-instrument diagnostics, Bayesian IV: notebook 11, and 11b on real Criteo data.



Reading the IV panel. Naive OLS (orange) is **precisely wrong**: a deceptively tight standard error around an estimate many standard errors away from the truth (the count is printed above, and it gets *worse* if you swap the clustered SE for the more “efficient” HC1 one — Step 0 prints both) — under the table’s routing rules it would have us wrongly *hold* prices fleet-wide, with great confidence. (Its naive-OLS point sits a touch below step 6’s endogenous hierarchical fit because this IV panel adds the cost-shift term — same story, slightly different panel.) 2SLS (blue) is **imprecisely right**: it lands within a couple of its (much wider) standard errors of the planted elasticity, because it discards every scrap of price variation except the cost-shock slice, which is clean by construction. That asymmetry is the whole IV bargain — the cure costs **variance**: real-world IV estimates are noisier and need bigger samples, and a *weak* instrument (first-stage $F \lesssim 10$) is worse than none, amplifying small exclusion violations into large bias. Managing that trade-off, with honest IV uncertainty in a Bayesian wrapper, is notebook 11’s whole job; notebook 11b repeats it on real Criteo data.

Other caveats, folded into the assumption ledger of step 3.

- **A4 · functional form.** Log-log assumes *constant* elasticity within region. We probed it in-sample (step 5’s curvature term — consistent with zero in this run, with the honest note that $\pm 5\%$ of price wiggle gives that test little power), but every P^* sits at or beyond the edge of the observed price band, where no in-sample check reaches. The staged-rollout advice of step 7 is the operational hedge, and elasticity should be re-estimated as the new prices generate data.
- **A3 · dynamics.** A $\text{lag}(\text{price})$ term would capture stockpiling / reference-price dynamics

(omitted here for clarity); on real scanner data always try it — pantry-loading masquerades as high short-run elasticity and vanishes at longer horizons.

- **Levels vs logs — a porting trap.** The model is for $\log Q$; if you ever forecast *units*, the lognormal mean needs the correction $\mathbb{E}[Q | P] = e^{\mu + \sigma^2/2}$, not e^μ . Our profit *uplift* is immune — it is a **ratio** of two expected profits with the same residual σ , so the $e^{\sigma^2/2}$ factor cancels in `uplift_pct` — but the moment you report absolute euros or units, it bites.
- **Pooling is a prior choice.** Too few regions or a mis-specified τ_β prior can over-pool — always inspect the fleet-vs-region spread (τ_β 's posterior), and remember step 4's non-centered reparametrization when the sampler starts complaining.

9 · The one-paragraph decision

To the CMO. Fleet demand is elastic — the fleet-mean elasticity lands close to -1.4 (exact value and credible interval in the dict below), with genuine region-to-region spread, so a single national price is leaving money on the table. For the majority of regions the posterior clears our 0.9 bar that demand is truly elastic ($\beta < -1$); for those we recommend moving toward the table's **P*** prices — the expected fleet profit uplift and its 90% interval are printed under the decision table — rolled out to a minority of stores first, because every P* sits at or beyond the price range we have ever observed. The remaining regions get **no price move**: their elasticity posterior straddles -1 , where the optimal-price formula is unstable, so they get a **controlled price test** — randomized $\pm 10\%$ price cells for 8 weeks, deliberately manufacturing the exogenous price variation (A2) that identification feeds on. Three conditionals we will not hide: **(i)** everything assumes prices were not historically managed against demand (A1) — if they were, step 8 says the true elasticity is *more negative* than we estimated, so the uplift is conservative on that axis; **(ii)** the constant-elasticity extrapolation (A4) cuts the other way — profit at prices we never charged is model, not measurement; **(iii)** the recommended prices scale one-for-one with the €8 marginal-cost figure, so a cost restatement rewrites the price list (though not the set-vs-test split). Act where the posterior is sure; measure where it is not.

```
{
"fleet_elasticity": -1.36,
"fleet_elasticity_ci90": [-1.54, -1.17],
"tau_beta": 0.3,
"coverage90_true_beta": 0.75,
"n_regions": 12,
"n_set_price": 8,
"n_controlled_test": 4,
"n_hold": 0,
"fleet_uplift_pct": 24.4,
"fleet_uplift_ci90_pct": [15.0, 35.3],
"marginal_cost_eur": 8.0,
"endogeneity_bias_at_kappa_1p2": 0.83,
"ovb_predicted_bias_at_kappa_1p2": 0.83,
"kappa_where_call_flips": 0.01,
"iv_first_stage_F": 27.2,
"beta_ols_endogenous": -0.63,
```

```
"beta_iv": -1.66,  
"beta_iv_se": 0.21,  
"true_fleet_elasticity_reference": -1.42  
}
```